

The empirical recurrence for OEIS A253483, recovered from its tail

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Abstract

OEIS A253483 records “Empirical recurrence of order 90” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 114$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 92$. Its last coefficient is $c_{90} = -197971148800 \neq 0$, so no further tail satisfies anything shorter; any order-90 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A253483 is “Number of $(2+1)X(n+1)$ 0..2 arrays with every $2X2$ subblock ne-sw antidiagonal difference unequal to its neighbors horizontally and nw+se diagonal sum unequal to its neighbors vertically.”. It has offset 1 and begins

558, 6804, 85464, 1069400, 13402934, 167888152, 2103506918, 26352981590, . . .

The entry states:

Empirical recurrence of order 90 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:03:53, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 114$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 114$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 90: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 90.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 228$ exact terms from $n = 2$ on, it returns order 90 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{90} c_i a(n-i),$$

c_1	4	c_{24}	−193150348757	c_{47}	7393873288030302	c_{70}	2208144246000464
c_2	108	c_{25}	25433497976	c_{48}	−6776065152693326	c_{71}	−587853822679176
c_3	112	c_{26}	231666666103	c_{49}	1572562238700212	c_{72}	−2156936028364240
c_4	−1463	c_{27}	−1067189655095	c_{50}	6956439116132747	c_{73}	2723580513969680
c_5	−937	c_{28}	1665848332539	c_{51}	−11491338375102019	c_{74}	−594708896536800
c_6	−4989	c_{29}	3099986717545	c_{52}	9206567520076156	c_{75}	338287250351136
c_7	−44919	c_{30}	−7758055553443	c_{53}	−3223043306264658	c_{76}	−1674093760954944
c_8	185505	c_{31}	4937061106106	c_{54}	−4206339834974659	c_{77}	1732124767243520
c_9	334107	c_{32}	2823959607598	c_{55}	4549642706617854	c_{78}	−108540056118144
c_{10}	−954129	c_{33}	−33065475723977	c_{56}	1488085291354540	c_{79}	−672327904354304
c_{11}	3218733	c_{34}	47119138026636	c_{57}	−3452143117661520	c_{80}	484537284107264
c_{12}	−5764723	c_{35}	−6076443281972	c_{58}	−1609087631175764	c_{81}	−70895048107008
c_{13}	−32130394	c_{36}	−18835657431123	c_{59}	13086142291797282	c_{82}	−70109054547968
c_{14}	84966474	c_{37}	132233828424934	c_{60}	−17260096418000151	c_{83}	62211067281408
c_{15}	−52588444	c_{38}	−349287349724392	c_{61}	5921216912469264	c_{84}	−18666323468288
c_{16}	−86669066	c_{39}	390694091532842	c_{62}	7544030261201788	c_{85}	447203966976
c_{17}	1201213149	c_{40}	−76962004999854	c_{63}	−11929611902694009	c_{86}	6519202643968
c_{18}	−2501424708	c_{41}	−939972253300770	c_{64}	7390092387289697	c_{87}	−5761951334400
c_{19}	−1600624533	c_{42}	2155383621558547	c_{65}	−1781583792220844	c_{88}	517711331328
c_{20}	8421159003	c_{43}	−2280751647987376	c_{66}	1523189181868974	c_{89}	797706289152
c_{21}	−18201871174	c_{44}	1249190348465390	c_{67}	−1965763918619856	c_{90}	−197971148800
c_{22}	25987425164	c_{45}	1188793746548497	c_{68}	−445794229442080		
c_{23}	69972921489	c_{46}	−4868709385041910	c_{69}	−489900977065076		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 92$, and it is the recurrence OEIS A253483 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{90} - c_1x^{90-1} - \dots - c_{90}$. Its constant term is $-c_{90} = 197971148800$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 90 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 90, so $\deg q = 90$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 90, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 90, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -197971148800 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-90 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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